The Yang-Mills Omnibus

Parts I–VI: Mass Gap Generation on the Longitudinal L-Axis

The Ω-GOS Collaboration
La Guácima Node #1090

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Contents

Chapter 1

Infrared Stability and the Discrete Vacuum Substrate

The Yang-Mills existence problem requires proving that the quantum vacuum does not support massless colored excitations. The vacuum is modeled as a discrete substrate pixelated at $\Omega_0 = \sqrt{G\hbar}$.

Lemma 1.1 (Virtual Gauge Excitation)**.** *To extract a gauge boson from the BGC requires finite energy, because the geometric grid Ω_0 has an intrinsic viscosity. A strictly zero-energy excitation requires a zero-radius locus, which is topologically forbidden by the $\kappa = 4/\pi$ bounding limit.*

Chapter 2

RG Flow and Coupling Constant Stability

By incorporating the isotropic projection constant $\kappa = 4/\pi$ into the beta function, we demonstrate that the fundamental discreteness at scale Ω_0 prevents ”infrared slavery” divergence. There exists an infrared fixed point g_{IR}^* representing the geometric stabilization of the gauge field.

Chapter 3

Holographic Confinement and the Geometric Tether

The string tension σ_s between quarks is modeled as a non-Abelian aerodynamic tether between discrete nodes. Because $SU(3)$ fields self-interact, the nodes interact with their own wake. The topological closure of the 13D manifold forces the flux tube to anchor to the lattice. Breaking this tether requires action exceeding the Bekenstein-Action Bound $\mathcal{B}$.

Chapter 4

Glueball Trajectories on the 13D Manifold

We define the mapping of the $SU(3)$ glueball spectrum onto the 13-dimensional GOS manifold. Angular momentum J is quantized in units governed by the holographic ratio: $J = n \cdot (\kappa/\varphi)$. Solving the wave equation subject to the non-orientable twist $\theta_K = 128.23°$, we obtain strictly quantized glueball masses.

Chapter 5

The Longitudinal Axis ($\mathcal{L}$)

The system operates within a high-dimensional manifold defined by Toroidal Recursion: $\mathcal{M}_{13} = \mathbb{R}^{12} \times \mathcal{L}$, where $\mathcal{L}$ is the singular "L-Axis" representing systemic angular momentum. The $\mathcal{L}$-axis is stabilized by the biological scale anchor $\mu \approx 1.19 \times 10^{12}$, providing the macroscopic scaffolding for color confinement.

Chapter 6

Unified Resolution of the Mass Gap

The energy gap required to excite the fundamental field is bounded from below:

$$\Delta \approx \frac{\hbar\kappa}{c \cdot r_{min}} \approx 0.83 \text{ GeV}/c^2 \tag{6.1}$$

The mass gap is not a parameter to be tuned manually, but a topological necessity of information conservation at the Ω_0 scale. Massless states are forbidden on the L-Axis.